

VISHNUCHARAN S

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PROFILE

Master's Student focusing on integrating AI and Robotics. Experienced in developing real-time pipelines for sensor fusion, learning-based perception, and machine vision. Dedicated to developing robust, real-time software solutions for autonomous robotic systems and intelligent industrial automation.

EDUCATION

- Master of Science in Robotic System Technology** Oct 2025 – Present
RWTH Aachen University Aachen, Germany
- **CGPA:** 1.6/4.0 (Up to I semester)
 - **Relevant Coursework:** Robotic Systems, Kinematics and Dynamics, ACDC, AI, RL, CV, Multi Body Dynamics
- Bachelor of Technology in Production Engineering** Sep 2021 – May 2025
National Institute of Technology, Tiruchirappalli (NIT Trichy) Tamil Nadu, India
- **CGPA:** 8.06/10 | **Minor:** Computer Science
 - **Relevant Coursework:** Industrial Robotics, Industry 4.0, Data Structures and Algorithms, Calculus, CAD

TECHNICAL SKILLS

Software & Engineering: Python, C++, SQL, Dev, MATLAB, SolidWorks, ANSYS, Creo, Simulink, Latex, Jupyter
Perception & AI: PyTorch, TensorFlow, CNNs, Transformers, OpenCV, Open3D, Ultralytics, Scikit-learn, NLTK
Robotics & Systems: ROS2, Gazebo, SLAM, Nav2, MoveIt2, Sensor Fusion, CUDA, Docker, Linux, Git
Visualization & Media: Matplotlib, RViz, Tableau, Power BI, Adobe Suite (Ps/Pr/Ae/Lr), Blender

RESEARCH EXPERIENCE

- Cross-Modal Depth Estimation for Traffic Light Detection [ACDC - RP]** Apr 2026 – Present
ika, RWTH Aachen University Aachen, Germany
- Developed geometric fusion of LiDAR-cameras using 4x4 transformation chains and perspective projection.
 - Engineered semantic frustum gating and median-Z filtering to isolate clusters & mitigate background edge-bleed.
- Welding Defect Detection in Metal Additive Manufacturing** Oct 2024 – May 2025
Advisor: Dr. V. Senthilkumar, NIT Trichy India
- Built a real-time defect detection using fine-tuned YOLOv12 Roboflow API for metal additive manufacturing.
 - Developed a Flask-based image/video inspection platform and an AI assistant with **84% model precision**.
- Text-Guided Image Segmentation** Oct 2023 – May 2024
Advisor: Dr. Sridevi M, NIT Trichy India
- Integrated CLIPSeg with FastSAM and optimized parameters for zero-shot text-guided semantic segmentation.
 - Improved multi-object segmentation reliability using adaptive thresholding and semantic prompt understanding.

PROJECTS

- Autonomous Warehouse Robot Simulation** May 2026
- Built a ROS2 + Gazebo mobile robot pipeline with SLAM-based mapping and AMCL localization using LiDAR.
 - Enabled Nav2-based autonomous navigation, classical CV object detection for goal-directed retrieval.
- ROS2-Based Multimodal Perception Pipeline for Sensor Reliability Monitoring** Apr 2026
- Built a ROS2-based multimodal pipeline fusing LiDAR (20Hz) and camera (14Hz) for real-time 3-class detection.
 - Used 12k+ rosbag to train a dual-branch ResNet18 late-fusion model and deployed ROS nodes with state-machine.
- Remote Sensing Image Captioner** Mar 2024
- Developed an image captioning system using InceptionV3 encoder and Transformer decoder for satellite data.
 - Achieved BLEU score of 0.59/1.00 (human: 0.60/1.00) and deployed a web app for real-time geo-satellite captioning.

LEADERSHIP & ACTIVITIES

- Director of Campus Affairs** | Technical Council, NIT Trichy 2022 – 2025
- Managed activities of 10+ technical clubs including Robotics Club and allocated a budget of Rs. 17 million.
 - Organized TransfiNITTe'24 and coordinated Pragyan, NIT Trichy's ISO-certified technical fest.
- Head of Design** | NITTFEST Cultural Fest 2023 – 2024
- Led 37-member media team producing digital content and mentored fest outreach campaigns.
- Community & Teaching**
- Mentored 25 underprivileged students for entrance exams ; worked as photographer for major institutional events.

ADDITIONAL INFORMATION

Languages: English (Proficient), German (A2-Learning), Tamil (Native), Hindi (Fluent)
Interests: Robotics, Perception & Sensing, Photography, Fitness & Sports, Music, Cinematography, Creative Tech
Soft Skills: Collaboration, Problem Solving, Adaptability, Technical Communication, Leadership, Service-oriented